

Predicting and Analyzing NBA Performance

Question 2: The Anatomy of Team Chemistry

What roster dynamics create team synergy, and does "sharing the ball" lead to success?

1. Introduction

"Team chemistry" is the most frequently cited, yet least mathematically defined, concept in basketball. When a team overachieves, analysts credit their chemistry. When a superteam fails, they blame egos. But what does chemistry actually look like in the data?

Initially, we set out to measure chemistry by looking at traditional on-court actions. Specifically, how evenly a team shares the ball. However, as the analysis deepened, the data forced us to shift our perspective. We discovered that on-court statistics alone cannot quantify synergy. To find true chemistry, we had to look beyond the court and analyze the human elements of roster construction: age, hierarchical structure, and the cultural diversity of the players themselves.

2. The Myth of "Sharing the Ball"

A common basketball narrative suggests that democratic offenses, where every player gets an equal share of the ball, are the pinnacle of team synergy. To test this empirically, we analyzed a dataset spanning 180 unique "team-seasons" (treating, for example, a franchise's 2018 roster as a separate observation from its 2019 roster). To establish a target variable for success, we assigned each team-season a **Composite Success Score**. This is a standardized metric combining three dimensions of performance: Net Rating (a team's point differential per 100 possessions, weighted at 45%), Playoff Wins (postseason success, 45%), and regular season Win Percentage (10%). Using this score, we isolated the extremes: the top 33% highest-performing team-seasons (61 team-seasons) against the bottom 33% lowest-performing (60 team-seasons). For both groups, we measured their Usage Variance. This is the standard deviation of the "Usage Percentage" (the proportion of team plays a specific player is involved in) among a team's core rotation players (the players who consistently get playing time). A low variance indicates a flat, democratic distribution of the ball, while a high variance indicates a strict hierarchy dominated by a single "alpha" star.

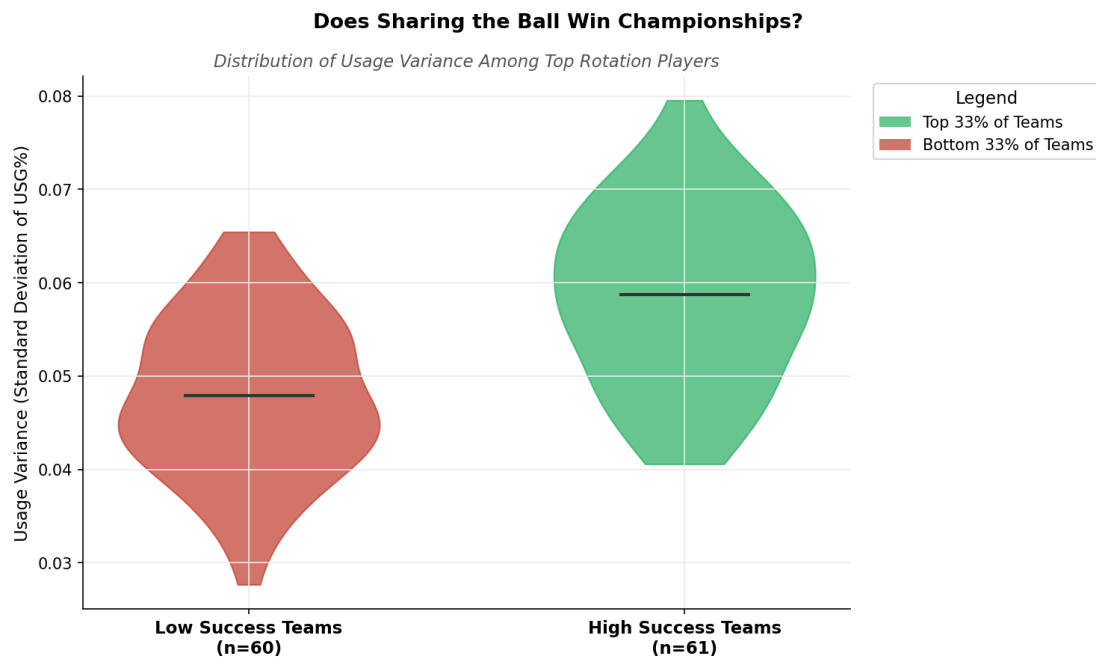


Figure 1. The distribution of Usage Variance across the most successful teams (Top 33%) versus the least successful teams (Bottom 33%).

Conclusion: The data immediately shatters the democratic ideal. As shown in the violin plot above, the most successful teams in the NBA operate with a significantly higher Usage Variance (averaging nearly 0.06) than bottom-tier teams (around 0.048). Winning basketball isn't about equal opportunity - it is about establishing a strict hierarchy where a primary star commands the offense while role players execute highly specific, limited jobs.

3. Defining Team Archetypes via Unsupervised Learning

Knowing that structural hierarchy (Usage Variance) correlates with winning, we wanted to identify the latent, natural structures of how front offices construct their rosters. To do this without domain bias, we applied an unsupervised machine learning algorithm, **K-Means clustering**, to all 180 team-seasons.

The algorithm clustered the data across three continuous features: Average Roster Age, Usage Variance (star concentration), and Age Variance (the mix of young prospects and older veterans). We set the algorithm to $k = 4$ to partition the league into four distinct structural quadrants. By comparing the mathematical centroids of these clusters to the global league averages, we identified four fundamental roster archetypes:

- **Veteran & Star-Centric:** Above-average age, high hierarchy.
- **Young & Star-Centric:** Below-average age, high hierarchy.

- **Veteran & Democratic:** Above-average age, flat hierarchy.
- **Young & Democratic:** Below-average age, flat hierarchy.

However, identifying these archetypes is only half the task: we needed to evaluate the efficacy of these structural models. To do this, we mapped each of the four machine-defined clusters against our target variable, the **Composite Success Score**, to determine which organizational philosophy actually wins games.

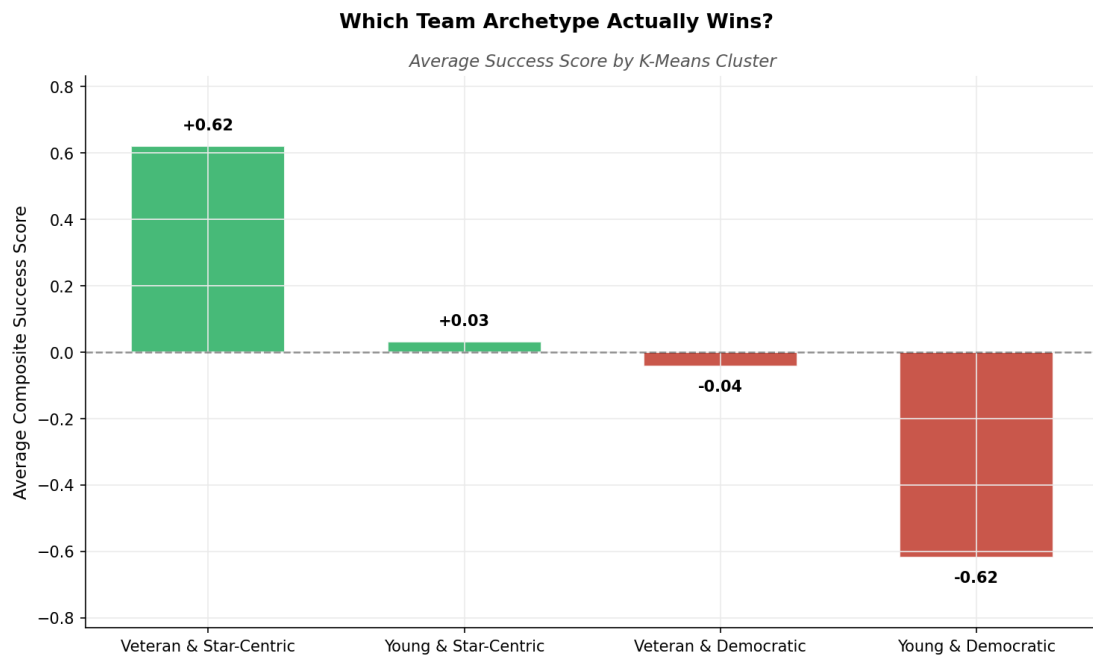


Figure 2. The average Composite Success Score (Wins, Net Rating, Playoff Wins) for each machine-defined cluster.

Conclusion: The results are stark. The "Veteran & Star-Centric" archetype vastly outperforms the rest of the league, averaging a +0.62 standard deviation in success. Conversely, the "Young & Democratic" model is a statistical disaster (-0.62). Building a roster of young players and instructing them to share the ball evenly is a mathematically proven recipe for failure.

4. The Cultural Synergy Hypothesis

Having established that rigid, star-centric hierarchies win games, we looked to investigate the actual human elements filling those roles. We extracted player birthplace data to evaluate whether the cultural composition of a roster affects how a team plays and whether they win.

International basketball is historically taught with a heavy emphasis on passing and team defense rather than individual isolation scoring. Our hypothesis: Injecting a higher concentration of international players into a roster should naturally drive up a team's democratic ball movement (measured by Assist Percentage: the percentage of teammate field goals a player assisted while on the floor).

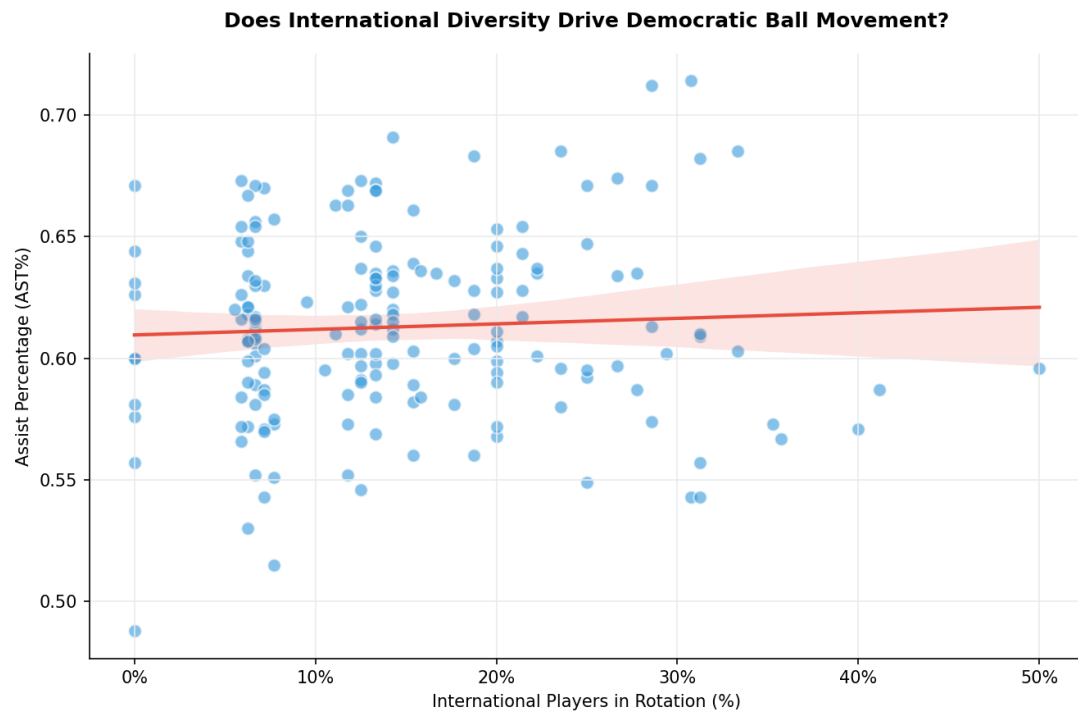


Figure 3. A regression analysis mapping the concentration of international players on a roster against the team's Assist Percentage.

Conclusion: Surprisingly, passports do not dictate passing. The completely flat regression line proves that a team with 0% international players can share the ball just as much as a roster with 30%. The overarching offensive system and the star hierarchy completely override a player's individual basketball upbringing.

Next, we tested if international diversity impacts winning, even if it doesn't change passing volume. To do this, we grouped the team-seasons into three distinct tiers based on the percentage of international players in their rotation. Because the NBA has historically been dominated by American players, these tiers were defined by the distribution of our dataset:

- **Low Diversity:** Less than 10% international players.
- **Medium Diversity:** 10% to 25% international players.
- **High Diversity:** Greater than 25% international players.

We then mapped these custom tiers against our target variable, the Average Composite Success Score.

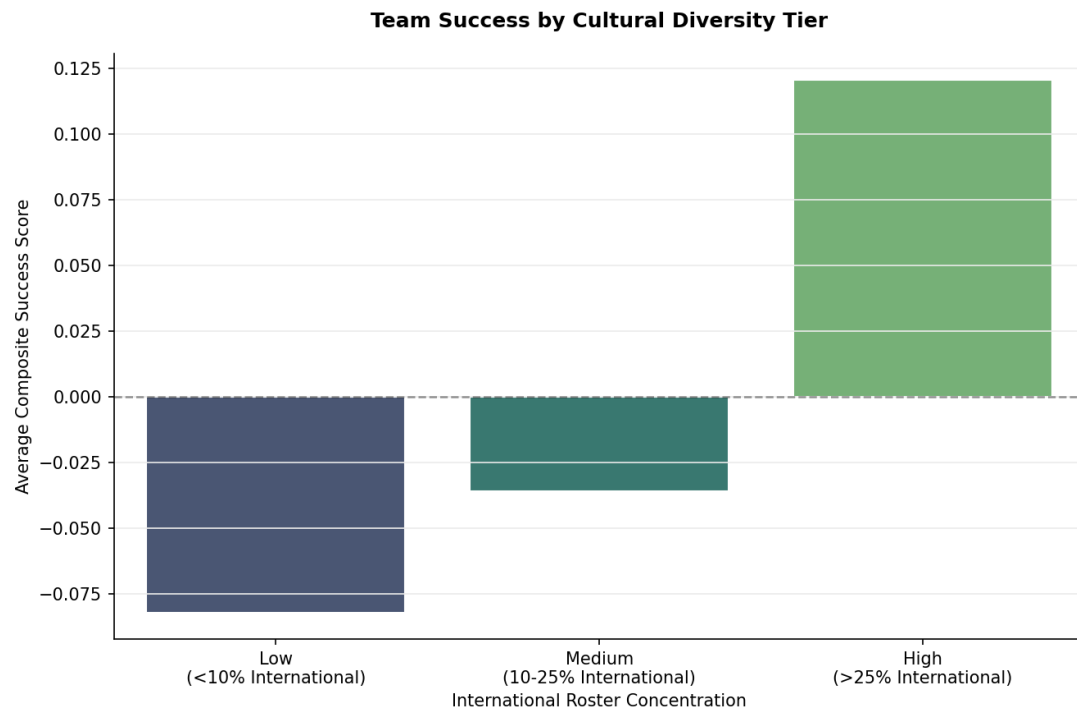


Figure 4. Average team success broken down by tiers of international roster concentration.

Conclusion: While international diversity does not change how often the ball is passed, it drastically changes the outcomes. Roster blending works. The "High" diversity tier (>25%) is the only group that produces a positive average success score (+0.120). By combining different basketball upbringings, front offices introduce a wider array of on-court problem-solving skills, making the collective unit more adaptable and resilient.

5. Predicting True Chemistry

Up to this point, we have measured general team success. However, raw success often just reflects a roster's raw physical talent, not their synergy. To isolate true synergy, we calculated a team's **Performance Delta** - the difference between their actual regular-season winning percentage and their expected winning percentage.

Expected winning percentage is calculated using the Pythagorean Expectation formula¹, which estimates how many games a team should have won based purely on their raw point differentials (Offensive and Defensive Ratings). A positive Performance Delta means the team won more games than their raw scoring stats suggest they should have - indicating that the team overachieved and played better than the sum of its parts.

We fed our structural and cultural traits into a Random Forest regressor to see which features hold the highest predictive weight in determining this overperformance.

¹ Originally developed for baseball by statistician Bill James and later adapted for the NBA, Pythagorean Expectation is a mathematical formula used to estimate a team's true baseline winning percentage. It filters out the "luck" or randomness of winning unusually close games to show how good a team *should* be based purely on their scoring output.

The formula relies on three components:

- **OFF_RTG (Offensive Rating):** An estimate of points scored per 100 possessions.
- **DEF_RTG (Defensive Rating):** An estimate of points allowed per 100 possessions.
- **The Exponent (13.91):** A mathematically derived constant specific to the NBA environment that optimally maps historical point differentials to actual win/loss records.

The formula:

$$\text{Expected W_PCT} = \frac{\text{OFF_RTG}^{13.91}}{\text{OFF_RTG}^{13.91} + \text{DEF_RTG}^{13.91}}$$

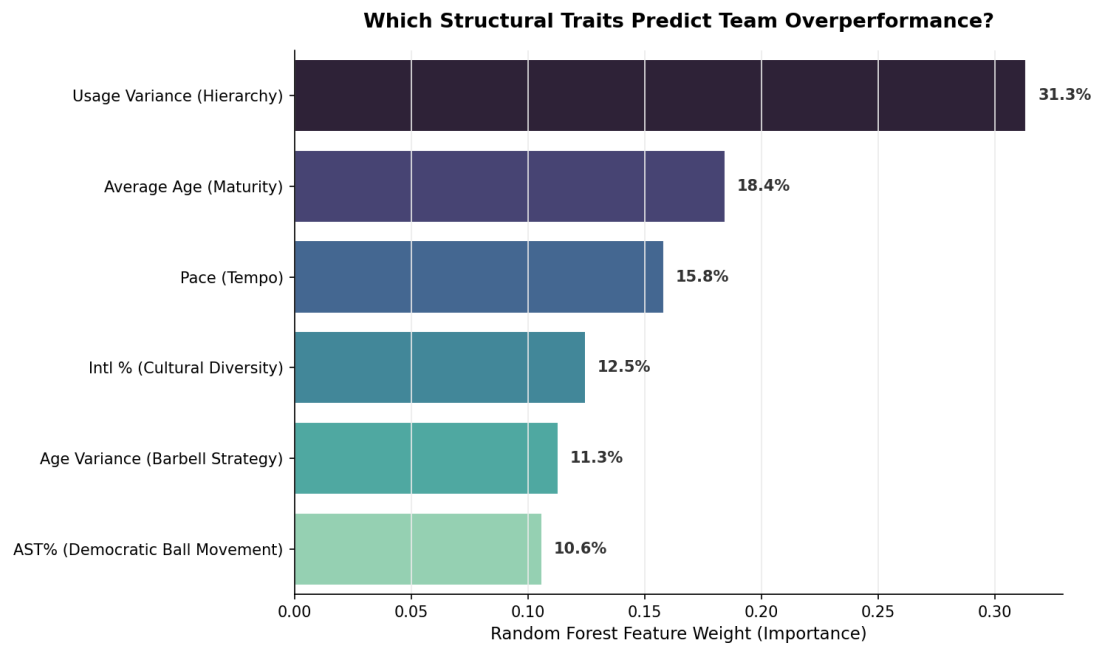


Figure 5. Random Forest feature importance for predicting a team's Chemistry Delta (over/underperformance).

Note on model variables: We initially hypothesized that the "Barbell Strategy" - mixing raw rookies with aging veterans (measured as Age Variance) - act as a synergy hack. However, the algorithm proved it statistically ineffective (11.3%). Additionally, while Pace (15.8%) registered as moderately important, we view it as a coaching tactical decision rather than an intrinsic measure of roster construction.

Conclusion: The machine learning model provides a definitive blueprint for roster synergy:

1. **Hierarchy (31.3%) is paramount:** Clear role definitions and a high Usage Variance are the strongest predictors of a team overachieving.
2. **Maturity (18.4%) over mixing:** Overachieving teams rely on a high average age. Barbell strategies are largely ineffective.
3. **Culture (12.5%) matters more than Passing (10.6%):** Cultural diversity is a marginally stronger driver of overperformance than democratic ball movement.

6. Final Q2 Conclusion

The data shatters the traditional narrative of team synergy. Good team-building is not about equal opportunity or passing the ball for the sake of passing. Whether assembling a championship basketball roster or building a highly autonomous software development unit in a new international market, the fundamental rules

of organizational synergy remain the same: overachieving collectives rely on absolute role clarity, collective maturity, and a blend of diverse backgrounds to solve problems efficiently.